Example. Evaluate the following integrals:

(i) $\displaystyle\int_0^2 \int_0^1 \int_0^2 x^2 + y^2 + z^2 \, dz \, dy \, dx$

(ii) $\displaystyle\int_0^4 \int_0^{3-3x} \int_0^{3-3x-y} 1 \, dz \, dy \, dx$

When studying double integrals, we found that they could be used to streamline the computation of a surface's average value and the area of a 2-dimensional region. Thus, it stands to reason that a *triple integral* can be used to help calculate the volume of 3-dimensional objects and the average value of a function on a 3-dimensional region. Such a 3-dimensional region is usually a closed and bounded region D:

Definition. Given a closed and bounded region D in 3-space,

$$\text{Volume of } D = \iiint\limits_D 1 \, dV = \iiint\limits_D 1 \, dz \, dy \, dx \, .$$

This definition makes sense intuitively, but the real tricky part is finding the limits of integration. Note that a higher-dimensional Fubini's theorem *does* hold, so the order '$dz \, dy \, dx$' isn't the only option—but it is often the most convenient to start with. In doing so,

Theorem. [Fubini's Theorem (Triple-Integral Form)] Let $F(x, y, z)$ be a continuous function on the closed and bounded region D defined by

$$a \leqslant x \leqslant b, \qquad g_1(x) \leqslant y \leqslant g_2(x), \qquad \text{and} \qquad f_1(x, y) \leqslant z \leqslant f_2(x, y),$$

where g_1, g_2, f_1, and f_2 are continuous. Then

$$\iiint\limits_D f(x, y, z) \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{f_1(x,y)}^{f_2(x,y)} F(x, y, z) \, dz \, dy \, dx.$$

Example. Setup and solve a triple integral that returns the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Example. Setup and solve a triple integral that returns the volume of the region D enclosed by the cylinder $z = 3y^2$ and the planes $x = 0$, $x = 4$, $y = -4$, and $y = 4$.

Example. Consider the closed and bounded tetrahedron D formed by connecting the points $(0,0,0)$, $(1,1,0)$, $(0,1,0)$, and $(0,1,1)$.

(i) Carefully draw and label a picture of D.

(ii) Setup a triple integral that returns the volume of D using the following order of integration:

(a) $dy\,dz\,dx$:

(b) $dz\,dy\,dx$:

We close with a final commentary on average value. It stands to reason that, given a function $F(x,y,z)$ defined on a closed and bounded region D,

$$\text{Average value of } F \text{ over } D = \frac{1}{\text{volume of } D} \iiint\limits_D F(x,y,z)\,dV$$